

On Optimizing Fault Coverage, Pattern Count, and ATPG Run Time Using A Hybrid Single-Capture Scheme for Testing Scan Designs

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Abstract

This paper presents a hybrid automatic test pattern generation (ATPG) technique using the staggered single-capture scheme followed by the one-hot single-capture scheme for detecting structural faults, which are neither timing-dependent nor sequence-dependent in a scan design. Structural faults are also called combinational faults or DC faults, such as stuck-at faults and bridging faults. Typically, the one-hot scheme achieves near maximum fault coverage, takes shorter ATPG run time, but produces a large pattern count, whereas the staggered scheme produces smaller pattern count but needs long ATPG run time and may suffer from some fault coverage loss. The proposed hybrid technique is intended to optimize fault coverage with respect to the one-hot scheme by exploring trade-offs between pattern count and ATPG run time of multimillion-gate scan designs. Experimental results show that the proposed hybrid technique can achieve higher fault coverage and up to 4X smaller pattern count than the one-hot scheme.

1. Introduction

Scan design is a *design-for-testability* (DFT) technique in which the storage elements in a sequential circuit are converted into scan cells and these scan cells are then stitched together to form scan chains during scan testing [1-4]. By reconfiguring all storage elements into scan cells, the complexity of *automatic test pattern generation* (ATPG) for sequential circuits is transformed into manageable ATPG for combinational circuits. Since the late 1990s, scan design has become the most widely used DFT technique.

In recent years, with shrinking device geometry due to advances in design and manufacturing technology, circuits containing millions or tens of millions of logic gates and tens of clock domains are common. One key challenge in ATPG is how to generate a minimum set of test patterns that achieves the highest fault coverage with the shortest ATPG run time.

ATPG effectiveness is highly dependent upon the capture-clocking scheme used. Traditionally, **one-hot clocking** is used to sequentially test every clock domain one by one. This scheme has the luxury of testing each clock domain by deactivating test clocks to other clock domains; however, it often only saves ATPG run time but generates many more test patterns than optimally required resulting in higher test cost.

In this paper, we only consider **structural faults**, such as stuck-at faults, bridging faults, and I_{DDQ} faults. We first discuss two other single-capture clocking schemes, namely, **simultaneous clocking** and **staggered clocking** [5-7], which can be used to remedy the problems encountered in one-hot clocking. We show that using the simultaneous clocking scheme alone can result in the smallest pattern count but may lead to significant fault

coverage loss. The staggered technique has shown to be effective in reducing pattern count [8-9] and achieving better N-detect fault coverage [10] but its ATPG time could be much longer. Lastly, we demonstrate that by using a hybrid scheme, which combines staggered clocking and one-hot clocking, one can optimize overall fault coverage, pattern count and ATPG time.

Most importantly, the proposed hybrid scheme (1) always achieves the same or higher fault coverage than the one-hot scheme alone, and (2) allows users to make a trade-off between pattern count and ATPG run time, without sacrificing the fault coverage of the design, in situations where staggered ATPG might take too long to finish. This hybrid scheme also addresses issues raised in [9] that it is uncertain whether the mixed clock-domain (one-hot clocking) and clock-concatenation (staggered clocking) approach could reach the one-hot fault coverage and whether it could complete the mixed ATPG run reaching the highest possible fault coverage within a specified run time for large designs.

This paper is organized as follows: Section 2 discusses various test timing control diagrams for detecting structural faults. Section 3 proposes the hybrid single-capture schemes. Section 4 shows results on two industrial designs, and Section 5 concludes.

2. Test Timing Control

An **intra-clock-domain structural fault** resides in one clock domain and gets detected within the same clock domain. An **inter-clock-domain structural fault** resides within a crossing clock domain and gets detected at the receiving clock domain.

Single-capture is a slow-speed test technique in which only one capture-clock pulse is applied to each clock domain. It is the simplest technique for testing all intra-clock-domain and inter-clock-domain structural faults. There are three capture-clocking approaches that can be used to implement the technique: (1) one-hot single-capture, (2) simultaneous single-capture, and (3) staggered single-capture. One major benefit of using these single-capture clocking schemes is that only a single, slow-speed *global scan enable* signal *GSE* is needed to drive each clock domain. This greatly simplifies physical implementation.

2.1. One-Hot Single-Capture

Using the **one-hot single-capture** approach, a capture pulse is applied to only one clock domain during each capture window, while all other test clocks are held inactive. An example timing diagram is shown in Figure 1. In the figure, since only one capture pulse (*C1* or *C2*) is applied during each capture window, this scheme can only test intra-clock-domain and inter-clock-domain structural faults. The main advantage of this approach is that the designer does not have to worry about clock skews between the clock domains since each clock domain is tested independently. The only requirement is that delays *d1* and *d2* be properly adjusted. Hence, this approach can be used for detecting structural faults. A major drawback is longer test time since all clock domains have to be tested one at a time.

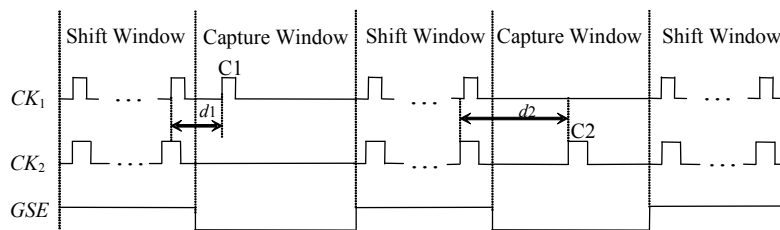


Figure 1. One-hot single-capture

2.2. Simultaneous Single-Capture

The long test time problem of one-hot single-capture can be resolved by using the simultaneous single-capture scheme illustrated in Figure 2. The **simultaneous single-capture** scheme allows testing to be performed on all clock domains in parallel. This scheme is quite helpful when signals in clock domains do not interact with each other. For clock domains where data may propagate from one clock domain to the other, the values of source scan cells in the originating clock domains will have to be forced to unknown values (X 's) during ATPG in order to avoid pattern mismatches.

The major advantage of this approach is that all intra-clock-domain structural faults can be tested simultaneously thus yielding much shorter ATPG time and pattern count. However, it exposes the design to two major drawbacks which are not present in one-hot single-capture: (1) the forced X 's on all source scan cells could cause significant fault coverage loss and (2) test compression may become very difficult with the need to mask off all X 's captured when using simultaneous clocking.

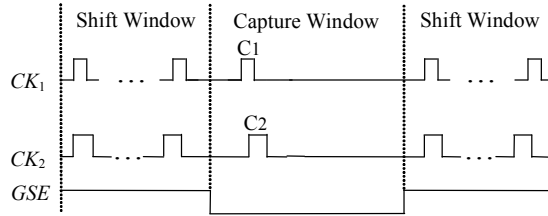


Figure 2. Simultaneous single-capture

2.3. Staggered Single-Capture

The fault coverage loss problem of simultaneous single-capture can be remedied using **staggered single-capture**. A test timing control example is shown in Figure 3. In this figure, capture pulses C₁ and C₂ are applied in a sequential or staggered order in the capture window to test all intra-clock-domain and inter-clock-domain structural faults in the two clock domains. If the two clock domains are synchronous to each other, then adjusting $d2$ will allow us to detect inter-clock-domain delay faults at-speed. In addition, since $d1$ and $d3$ can be adjusted to be as long as desired, a single, slow-speed global scan enable signal GSE can be used. This significantly simplifies physical implementation for designs with multiple clock domains. However, there may be some structural fault coverage loss among clock domains if an ordered sequence of capture clocks is used across all capture cycles. This fault coverage loss is mostly related to sequentially redundant faults that can only be detected when one-hot clocking is employed.

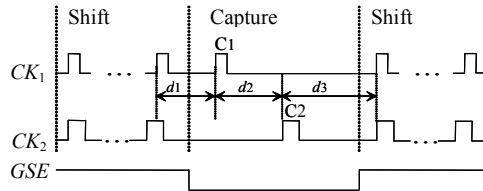


Figure 3. Staggered single-capture

3. Hybrid ATPG Techniques

This section describes two hybrid capture-clocking schemes that were patented in [11] and [12]. The proposed hybrid scheme combines two single-capture approaches, based on the result of clock grouping that identifies independent or non-interacting clock domains. The corresponding ATPG flow will also be described.

3.1. Clock Grouping

An effective technique for reducing ATPG time of a scan design is to first identify all clock domains that do not interact with each other. This is achieved by conducting **clock grouping** that analyzes all data paths in the circuit in order to identify all independent or non-interacting clocks. These clocks are then grouped together and applied simultaneously. If a data path originates at a clock domain and terminates at another clock domain, the two clocks controlling the data path need to be placed in different clock groups.

Independent clock groups identified by clock grouping can then be used for capture-clocking using the three basic single-capture schemes described in the previous section. Since all of the above-mentioned clocking schemes may result in fault coverage loss, large pattern count, and/or long ATPG time, we describe two hybrid schemes in the following for users to make a trade-off between pattern count and ATPG time while maintaining the same fault coverage achievable by the one-hot clocking scheme.

3.2 Simultaneous-Followed-By-One-Hot ATPG

One hybrid ATPG approach is to apply the simultaneous single-capture clocking scheme for all clock domains in the first phase and the one-hot single-capture clocking scheme in the second phase. In the first phase, all clocks are treated as a single clock. In practical applications, data can propagate from one clock domain to another clock domain in a predetermined clocking order. To avoid clock skew among interacting clock domains, one solution is to force all fanout branches of each originating flip-flop in an originating clock domain to unknown (X) values. Since all clocks are treated as a single clock, there is no need to perform clock grouping when the simultaneous scheme is employed. In the second phase, all remaining faults that are not detected in the first phase are targeted for one-hot ATPG. These remaining faults must include all faults from the first phase that are accidentally marked as untestable or undetected due to the existence of X 's.

Since one-hot clocking is used in the second phase, this hybrid approach can achieve the same fault coverage as one-hot clocking alone but the hybrid approach often can generate fewer test patterns with shorter overall ATPG time when compared with the one-hot only approach. In addition, although clock grouping is not required in the first phase, performing one-hot ATPG based on clock groups rather than individual clocks, in general, can reap additional benefits of smaller pattern count and shorter ATPG time.

3.3. Staggered-Followed-By-One-Hot ATPG

Although above-mentioned hybrid approaches can lead to better results than using the one-hot approach alone, it may suffer from the same X -masking issue ATPG compression faces. If test response compaction logic in the circuit cannot effectively tolerate massive X 's created by simultaneous clocking, a significant fault coverage loss may happen.

In order to solve this problem, another hybrid approach is to apply the staggered single-capture scheme in the first phase and the one-hot single-capture scheme in the second

phase. In the first phase, all clock groups are specified in a predetermined, sequential or staggered order. ATPG is then conducted based on the staggered order. To reduce ATPG time, circuit model expansion based on the ordered sequences of clock groups is performed on the scan design during preprocessing. Since the staggered clocking scheme specifically deploys physically disjoint capture clock pulses from different clock domains (in our case, different clock groups), there is no need to insert X 's at the fanout branches of each originating flip-flops in any originating clock domain. Therefore, the staggered approach will not create unnecessary X 's that complicate test response compaction in a compression design. Since staggered clocking can cause the ATPG program to mark hard-detected faults as untestable or undetected due to the ordered sequence of clock groups, the second phase running one-hot ATPG is required to re-target those faults.

3.4 ATPG Flow

ATPG flow of the staggered-followed-by-one-hot scheme is shown in Figure 4 and summarized as follows: Starting with a scan-stitched netlist, clock grouping analysis is first conducted. With available clock grouping and capture order information, the ATPG program transforms the scan-stitched netlist into a combinational circuit ready for test pattern generation. Remodeling, learning, implication and other pre-processing steps are then conducted. During ATPG, switch criteria are closely monitored for the program to switch over from staggered ATPG to one-hot ATPG.

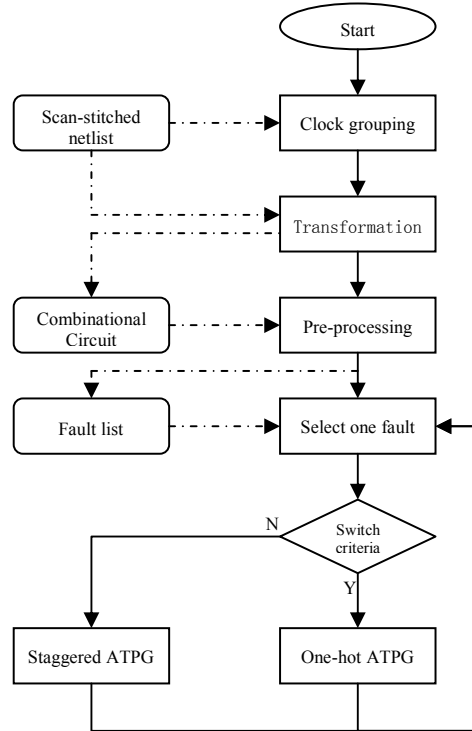


Figure 4. Hybrid ATPG flow chart

There are different ways to set the switch-over criteria such as certain increments in fault coverage over pattern count or a pre-determined amount of run time. For this paper, we

define the switch-over criterion, called **switch ratio**, as the percentage of faults that have been processed during the entire ATPG process over total faults.

4. Experimental Results

The proposed hybrid staggered-followed-by-one-hot single-capture scheme has been applied to many industrial designs. We present the results on two multimillion-gate designs to illustrate the effectiveness of the proposed scheme.

4.1 Design Statistics

Table 1 summarizes the statistics of two industrial designs A and B. We developed a program to identify all independent clock groups. Because clock domains in the same clock group do not interact with one other, they can be enabled simultaneously during capture without adverse effect from clock skews.

Table 1. Design statistics

	<i>Design A</i>	<i>Design B</i>
# of Gates	1.2M	4.9M
# of Faults	3,954,048	11,395,890
# of Flip-Flops	105K	327K
# of Clock Domains	38	10
# of Clock Groups	7	8

4.2 Results

Stand-alone one-hot single-capture and staggered single-capture clocking schemes were first applied independently to both designs A and B. The platform used was a 2.5-GHz 64-bit PC running the Linux operating system. Table 2 summarizes the experimental results. Single stuck-at fault model was assumed in ATPG. The one-hot results were obtained by running one clock at a time, while the staggered results were obtained by running all clock groups in a given staggered order.

Table 2. Experimental Results on Designs A and B

	<i>Design A One-Hot</i>	<i>Design A Staggered</i>	<i>Design B One-Hot</i>	<i>Design B Staggered</i>
Hard-Detected Faults	3,650,163	3,639,949	10,432,447	10,442,183
Fault Coverage (%)	92.58%	92.32%	92.42%	92.51%
Pattern Count (one-hot / staggered)	6,339	1,464 (4.33X)	2,591	1,483 (1.75X)
ATPG Run Time	1:17:45	0:57:46	2:49:11	6:31:59
Memory Usage	1.6GB	2.0GB	6.0GB	7.2GB

The results show that one-hot clocking demands less memory than staggered clocking. At the cost of larger pattern count, one-hot clocking should have higher fault coverage and shorter ATPG run time than staggered clocking; however, higher fault coverage and shorter ATPG run were observed for staggered clocking for Designs B and A, respectively. The reason for the fault coverage increase may be because faults aborted in the one-hot clocking scheme (due to backtrack limit) got detected by patterns targeting other faults using the

staggered clocking scheme. The reason for reduced run time may be due to staggered clocking proving many sequentially redundant faults, detectable only by one-hot clocking.

It is also interesting to note that, on average, ATPG time for stuck-at faults spent on Designs A and B using the staggered clocking scheme is quite linear, approximately one-hour per one million gates. This implies that ATPG on a 20-million gate design may take less than one day.

The results using the proposed hybrid clocking scheme on Designs A and B are listed in Tables 3 and 4, respectively. In our experiments, staggered clocking is switched to one-hot clocking after a certain percentage of faults have been processed. The switch ratio is shown in the first column of each table. In the other four columns, hard-detected faults, fault coverage, pattern count, and ATPG time are shown as $X + Y (Z)$. The first number, X , is the result using staggered clocking; the second number, Y , is from using one-hot clocking; and the third number, Z , is the sum of X and Y . The results on both tables show that as the switch ratio increases, the pattern count decreases and fault coverage increases. ATPG run time, generally speaking, increases except in the case of 95% switch ratio.

Table 3 further shows the effectiveness of the hybrid clocking scheme. First, the hybrid fault coverage is always higher than one-hot clocking alone. Second, the pattern count from using one-hot clocking is 3.70 ($= 6339 / 1711$) times the pattern count using the hybrid approach when the switch ratio is 100%. Lastly, as shown in the last row of the table, the staggered clocking can already achieve 83.99% fault coverage with merely 85 patterns; this is crucial when ATE memory size is a concern.

Table 3. Hybrid Clocking Results on Design A

<i>Switch Ratio</i>	<i>Hard-Detected Faults</i>	<i>Fault Coverage (%)</i>	<i>Pattern Count</i>	<i>ATPG Time</i>
100%	3,639,949 + 22,209 (3,662,158)	92.32% + 0.56% (92.88%)	1,464 + 247 (1,711) (3.70X)	0:57:46 + 0:12:42 (1:10:28)
99%	3,626,520 + 34,658 (3,661,178)	91.98% + 0.88% (92.86%)	848 + 1,028 (1,876) (3.38X)	0:36:46 + 0:19:45 (0:56:31)
95%	3,483,666 + 171,026 (3,654,692)	88.35% + 4.34% (92.69%)	201 + 2,939 (3,140) (2.02X)	0:08:38 + 0:36:55 (0:45:33)
90%	3,311,452 + 342,690 (3,654,142)	83.99% + 8.69% (92.68%)	85 + 3,533 (3,618) (1.75X)	0:05:45 + 0:41:08 (0:46:53)

Table 4 also shows the effectiveness of the hybrid clocking scheme on Design B. At each switch ratio, the hybrid scheme generates fewer test patterns and requires longer ATPG run time than the one-hot clocking scheme given in Table 2. Similar to Design A, the reduction in pattern count seems to fall within 20% when the switch ratio is 90% or lower. For

Design B, the fault coverage of the hybrid clocking scheme at each switch ratio is also found to be slightly higher than that of one-hot clocking scheme. These results are consistent with Design A.

Table 4. Hybrid Clocking Results on Design B

<i>Switch Ratio</i>	<i>Hard-Detected Faults</i>	<i>Fault Coverage (%)</i>	<i>Pattern Count</i>	<i>ATPG Time</i>
100%	10,442,183 + 651 (10,442,834)	92.51% + 0% (92.51%)	1,483 + 43 (1,526) (1.69X)	6:31:59 + 1:03:22 (7:35:21)
95%	10,119,541 + 318,037 (10,437,578)	89.65% + 2.82% (92.47%)	381 + 1,488 (1,869) (1.38X)	0:58:21 + 2:07:39 (3:06:00)
90%	9,572,488 + 863,072 (10,435,560)	84.80% + 7.65% (92.45%)	158 + 2,019 (2,177) (1.19X)	0:34:22 + 2:16:35 (2:50:57)

For Design A, Table 5 shows the difference between using one-hot and staggered clocking with 38 individual clocks *versus* 7 clock groups, respectively. The results indicate that performing ATPG based on clock grouping leads to much smaller pattern count and run time, but not necessarily higher fault coverage, in both one-hot and staggered cases.

Table 5. Experimental Results on Design A

	<i>One-Hot 38 clocks</i>	<i>One-Hot 7 clock groups</i>	<i>Staggered 38 clocks</i>	<i>Staggered 7 clock groups</i>
Hard-Detected Faults	3,650,163	3,653,526	3,641,227	3,639,949
Fault Coverage (%)	92.58%	92.66%	92.35%	92.32%
Pattern Count (38 clocks / 7 clocks)	6,339	3,982 (1.59X)	1,483	1,464 (1.01X)
ATPG Run Time	1:17:45	0:47:32	2:37:59	0:57:46

In summary, the experimental results on multimillion-gate designs show that (1) clock grouping effectively reduces pattern count and (2) hybrid clocking, on average, can result in 1.2X to 4X reduction in pattern count and slightly higher fault coverage than one-hot clocking alone. One-hot clocking, however, does have the benefit of shorter ATPG run time. Therefore, we recommend using one-hot clocking at an early development stage to predict the fault coverage and pattern count of a design. By the time the design is being taped out, one should apply hybrid clocking to reduce pattern count and improve the circuit's fault coverage to near maximum.

5. Conclusions

Scan designs containing multimillion logic gates are common today. When a scan design contains asynchronous clock domains, using the conventional one-hot and simultaneous single-capture schemes for capture clocking poses serious challenges. The drawback of the one-hot scheme is large pattern count, which leads to long test time and high test cost. The simultaneous scheme could reduce pattern count; however, the fault coverage loss and the negative impact on test compression may render the approach unacceptable.

This paper presents a hybrid *automatic test pattern generation* (ATPG) technique that combines the staggered single-capture and one-hot single-capture schemes to optimize fault coverage, pattern count, and ATPG run time for large scan designs. Experimental results on two industrial designs have demonstrated the effectiveness of the proposed scheme. When employed alone, the staggered scheme can drastically reduce pattern count at the expense of long ATPG time. In the hybrid scheme, ATPG switches from the staggered scheme to the one-hot scheme after a specified percentage of faults have been processed – this reduces ATPG run time at the cost of slight increase in pattern count. Since the one-hot scheme is applied after the staggered scheme, the hybrid scheme theoretically will achieve the same fault coverage as the one-hot scheme alone. In practice, however, we have observed higher fault coverage in all cases.

This paper only discusses capture-clocking schemes for detecting structural faults. While good results have been obtained, further investigation on how to automatically select the switch ratio remains to be explored. We plan to extend the hybrid scheme for detecting delay faults in scan designs. Through the experiments, we hope a near optimal solution among the three key ATPG parameters – fault coverage, pattern count and ATPG run time – could be empirically established.

6. References

- [1] M. L. Bushnell and V. D. Agrawal, *Essentials of Electronic Testing for Digital, Memory & Mixed-Signal VLSI Circuits*, Springer, Boston, 2000.
- [2] N. K. Jha and S. K. Gupta, *Testing of Digital Systems*, Cambridge University Press, London, 2003.
- [3] L.-T. Wang, C.-W. Wu, and X. Wen, Eds., *VLSI Test Principles and Architectures: Design for Testability*, Morgan Kaufmann, San Francisco, 2006.
- [4] L.-T. Wang, C. E. Stroud, and N. A. Touba, Eds., *System-on-Chip Test Architectures: Nanometer Design for Testability*, Morgan Kaufmann, San Francisco, 2007.
- [5] R. Apte, “Cutting SoC Test Costs with the Right Kind of Scan,” *EDAVision Magazine*, February Issue, 2002 <http://www.edavision.com/vision.php?article=200202/tool.html>.
- [6] L.-T. Wang, P.-C. Hsu, and X. Wen, “Multiple-Capture DFT System for Detecting or Locating Crossing Clock-Domain Faults During Scan-Test,” United States Patent No. 7,260,756, August 21, 2007.
- [7] SynTest Technologies, “ATPG User’s Manuals – TurboScanTM and VirtualScanTM,” SynTest Technologies, Inc., Sunnyvale, CA, 2008.
- [8] V. Jain and J. Waicukauski, “Scan Test Data Volume Reduction in Multi-Clocked Designs with Safe Capture Technique,” in *Proc. IEEE Int. Test Conf.*, pp. 148-153, October 2002.
- [9] X. Lin and R. Thompson, “Test Generation for Designs with Multiple Clocks,” in *Proc. ACM/IEEE Design Automation Conf.*, pp. 662-667, June 2003.
- [10] G. Bhargava, D. Meehl, and J. Sage, “Achieving Serendipitous N-detect Mark-Offs in Multi-Capture-Clock Scan Patterns,” in *Proc. IEEE Int. Test Conf.*, Paper 30.2, October 2007.
- [11] L.-T. Wang, K. S. Abdel-Hafez, X. Wen, B. Sheu, and S.-M. Wang, “Smart Capture for ATPG (Automatic Test Pattern Generation) and Fault Simulation of Scan-Based Integrated Circuits,” United States Patent No. 7,124,342, October 17, 2006.
- [12] K. S. Abdel-Hafez, L.-T. Wang, B. Sheu, Z. Wang, and Z. Jiang, “Method for Performing ATPG and Fault Simulation in a Scan-Based Integrated Circuit,” United States Patent No. 7,210,082, April 24, 2007.